Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Garland, Jay L Ph.D.		
Project Title:	Treatment of Urine/Condensate Spacecraft Wastewater Using Hollow Fiber Membrane Biofilm Redox Control Reactor		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32899	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	10/01/2003	**End Date:**	09/30/2005
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Original end date was 10/1/2006.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			
	This Pilot Study Project proposes to develop and evaluate a prototype Membrane Biofilm Redox Control Reactor for treatment and reclamation of wastewater for space flight applications. The prototype reactor will employ hollow fiber modules to deliver gaseous electron acceptors and donors to attached microbial biofilms. The use of hollow fibers provides a high surface area per reactor volume for gas transfer and biofilm attachment, resulting in high volumetric reaction rates. Hollow fiber membranes also provide a bubbleless gas supply suitable for gravity free operation in space. The goal of the proposed pilot study is to demonstrate a high degree of removal of nitrogen and organic compounds from an analog of the current ISS wastewater stream (urine and atmospheric condensate) in a single biofilm reactor. Following the successful demonstration of the technology, a Rapid Technology Development Team will be assembled to define a biological solution to an immediate Advanced Life Support requirement (i.e., water reclamation on ISS).		

Task Description:	Typical terrestrial approaches for removal of nitrogen and organic compounds from wastewater rely on coupling nitrification, an aerobic, autotrophic process that converts ammonia to nitrate, with denitrification, an anaerobic heterotrophic process that converts nitrate to N2 gas and is fueled by organic compounds in the wastewater. Spacecraft wastewater, particularly in near term missions without hygiene water, lacks sufficient organic material for complete denitrification. In spacecraft water recovery, the resulting residual nitrate would be an extra load on downstream processing systems such as reverse osmosis membranes. Total nitrogen removal from current ISS wastewater will require anaerobic ammonia oxidation or denitrification fueled by an exogenous source of electron donor or organic carbon. A major component of the pilot study is to evaluate the effects of the supply strategy of oxygen and reducing gas on the desired biochemical reactions and total nitrogen removal, and to delineate the key effects of the redox environments within the reactor on overall reactor performance. To evaluate anaerobic ammonia oxidation in the hollow fiber bioreactor, a continuous, limited oxygen supply strategy will be used to establish partial nitrification to nitrite and oxidation of ammonia to dinitrogen using the nitrite. Denitrification enhancement will be evaluated by oxygen on/off strategies, the use of non-oxygenated microbial attachment surfaces, and through addition of molecular hydrogen to promote autohydrogenotrophic denitrification. The prototype will use two separate, independently controlled gas supply systems to provide air or pure oxygen, and optionally molecular hydrogen. The two systems will each consist of parallel hollow fiber strands axially oriented in a column bioreactor. The individual hollow fiber strands of the two systems will be alternating and interlaced along the reactor column. The use of two gas supply systems will enable the separate supply of air, or oxygen, through one series of strands, and the optional supply of electron donor (hydrogen gas) through the other interlaced set of hollow fiber strands. The dual design will provide a high degree of operational flexibility by enabling the supply of electron acceptor and donor to be uncoupled in space and time. A systematic experimental evaluation will be performed of the effects of gas supply on redox environments, microbial transformations, and overall reactor performance in meeting the treatment goals. The strategies for the timing and intensity of gas supply will be adaptively evolved in a systematic fashion to delineate the critical methods and factors needed to achieve treatment objectives. A preliminary evaluation will be conducted of the membrane bioreactor product water quality to characterize its pretreatment requirements for the reserve-osmosis treatment that will be employed for water recovery in space flight applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated: 02/27/2004)